

DS 2003: Communicating with Data

Computer Setup Guide

Anaconda · Conda Environments · VS Code

This guide walks through everything from Lecture 2: installing Anaconda, building the conda environment for this class, installing VS Code, and configuring VS Code to use that environment.

Steps are given separately for Windows and Mac wherever they differ.

By the end of this guide, you'll have:

- ✓ Anaconda installed
- ✓ A **DS2003_F26** conda environment with the packages this class needs
- ✓ VS Code installed, with the Python and Jupyter extensions
- ✓ VS Code's interpreter set to your **DS2003_F26** environment

1. Install Anaconda

Anaconda is the supplier that includes Python and the scientific libraries you'll use all semester (NumPy, pandas, matplotlib, and more), plus **conda**, the tool you'll use to manage them. If you already have Anaconda installed, skip to Section 2.

WINDOWS

- 1 Go to **anaconda.com/download** and download the Windows installer (64-bit).
- 2 Run the downloaded **.exe** file and click through the installer using the default options.
- 3 On the “Advanced Installation Options” screen, you can leave both checkboxes (including “Add Anaconda to my PATH environment variable”) **unchecked** — the installer warns against it, and you won't need it. Use the **Anaconda Prompt** (installed automatically) any time you need a terminal with conda available.
- 4 Once installation finishes, open **Anaconda Prompt** from the Start menu and run `conda --version` to confirm it installed correctly.

MAC

- 1 Go to **anaconda.com/download** and download the macOS installer — choose **Apple Silicon** if your Mac has an M-series chip (M1/M2/M3/M4), or **Intel** for an older Mac. (Apple menu → About This Mac tells you which you have.)
- 2 Open the downloaded **.pkg** file and click through the graphical installer using the default options.
- 3 Open a new **Terminal** window (Spotlight search → “Terminal”) and run `conda --version` to confirm it installed correctly. If Terminal doesn't recognize conda, close and reopen Terminal — the installer needs a fresh shell to take effect.

2. Create Your Class Conda Environment

A conda **environment** is an isolated set of a Python version plus packages — think of it as its own kitchen with its own pantry, so the packages for this class never conflict with anything else on your computer. You'll create one environment, named **DS2003_F26**, and use it for everything in this course.

WINDOWS

- 1 Open **Anaconda Prompt** from the Start menu (not the regular Command Prompt).

MAC

- 1 Open **Terminal**.

From here the command is identical on both operating systems. Run:

```
conda create --name DS2003_F26 python=3.12 numpy pandas scipy ipykernel  
matplotlib
```

Type y and press Enter when prompted to confirm the package list.

What this installs: Python 3.12, plus NumPy, pandas, SciPy, matplotlib, and **ipykernel** (which lets this environment show up as a notebook kernel in VS Code / Jupyter). You can always add more packages later with `pip install <package>` once the environment is activated.

Once it finishes, verify the environment exists and can be activated:

```
conda activate DS2003_F26
```

Your terminal prompt should now show (DS2003_F26) at the start of the line instead of (base).

3. Install VS Code

VS Code is the editor (“the restaurant”) where you’ll write code, view data files, and run everything through a built-in terminal.

WINDOWS

- 1 Go to **code.visualstudio.com** and download the Windows installer.
- 2 Run the installer. On the “Select Additional Tasks” screen, it’s worth checking “**Add to PATH**” (usually checked by default).
- 3 **Important, Windows only:** the first few times you open VS Code, open it **through Anaconda Navigator** rather than from the Start menu directly — open **Anaconda Navigator**, find the **VS Code** tile, and click **Launch**. This makes sure VS Code can see your conda installation right away. Once your interpreter is set (Section 5), opening VS Code normally works fine too.

MAC

- 1 Go to **code.visualstudio.com** and download the macOS installer (choose Apple Silicon or Intel, matching your choice in Section 1).
- 2 Unzip the download and drag **Visual Studio Code.app** into your **Applications** folder.
- 3 Open VS Code from Applications. The first time, macOS may show a security prompt since it was downloaded from the internet — click **Open** to allow it.

4. Install VS Code Extensions

The same steps work on both Windows and Mac.

- 1 In VS Code, click the **Extensions** icon in the left-hand sidebar (it looks like four squares), or press `Ctrl+Shift+X` (Windows) / `Cmd+Shift+X` (Mac).
- 2 Search for **Python** (published by Microsoft) and click **Install**.
- 3 Search for **Jupyter** (published by Microsoft) and click **Install**. This is what lets VS Code open and run `.ipynb` notebook files directly.
- 4 *Optional:* search for **CSV** (e.g., “Rainbow CSV”) if you'd like a readable way to preview `.csv` data files without opening them as a notebook.

5. Set the Python Interpreter in VS Code

Why this step matters: selecting the interpreter through the Command Palette configures VS Code's entire experience — linting, debugging, the Run button, and any new terminal you open — to use the **DS2003_F26** environment. Activating an environment by typing `conda activate` directly into a terminal only affects **that one terminal session**. Do the Command Palette version once per project; you generally won't need to think about it again.

WINDOWS

- 1 Open your class folder in VS Code (**File** → **Open Folder...**).
- 2 Open the Command Palette with `Ctrl+Shift+P`.

MAC

- 1 Open your class folder in VS Code (**File** → **Open Folder...**).
- 2 Open the Command Palette with `Cmd+Shift+P`.

From here the remaining steps are identical:

- 3 Type **Python: Select Interpreter** and press Enter.
- 4 Choose the entry that shows **DS2003_F26** (it will be listed as a conda environment).
- 5 Open (or create) a `.ipynb` notebook and confirm **DS2003_F26** now appears in the top-right corner of the notebook, or in the bottom status bar for a `.py` file.

6. Verify Your Setup

Open a new Jupyter notebook in VS Code, make sure a code cell contains the following, and run it:

```
import pandas as pd
import matplotlib.pyplot as plt
print("Setup complete!")
```

If it runs without errors and prints `Setup complete!`, you're ready for lab. If you see a `ModuleNotFoundError`, double check that the interpreter shown for the notebook is **DS2003_F26** (Section 5) and that you installed the packages in Section 2 into that same environment.

7. Troubleshooting

Everyone's computer is set up a little differently, so don't be surprised if something here doesn't go exactly as written. A few common snags:

WINDOWS

- `'conda'` is not recognized — use **Anaconda Prompt** instead of the regular Command Prompt, which doesn't know about conda by default.

MAC

- `conda: command not found` — close and reopen Terminal after installing Anaconda; if it still doesn't work, run `conda init zsh` (or `conda init bash` if that's your shell) and then reopen Terminal again.

Both platforms:

- **DS2003_F26 doesn't show up in “Python: Select Interpreter”** — reload the VS Code window (Command Palette → **Developer: Reload Window**) after creating the environment.
- **Still stuck?** Come to office hours or post on Piazza — don't spend hours fighting your computer alone the week before a lab is due.